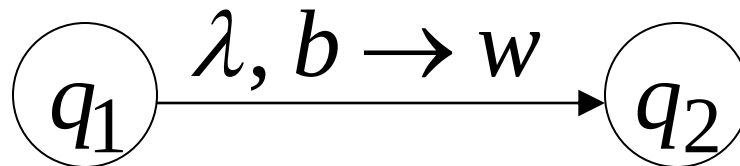
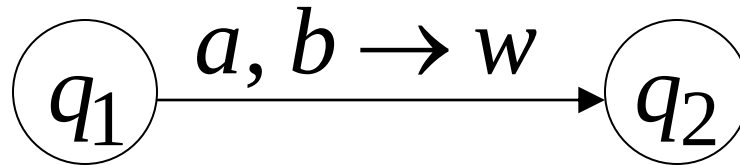


DPDA

Deterministic PDA

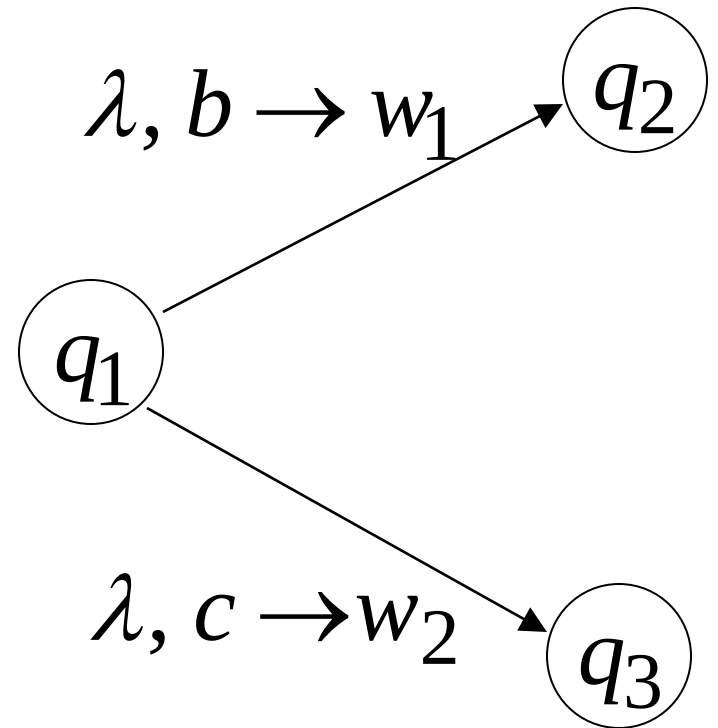
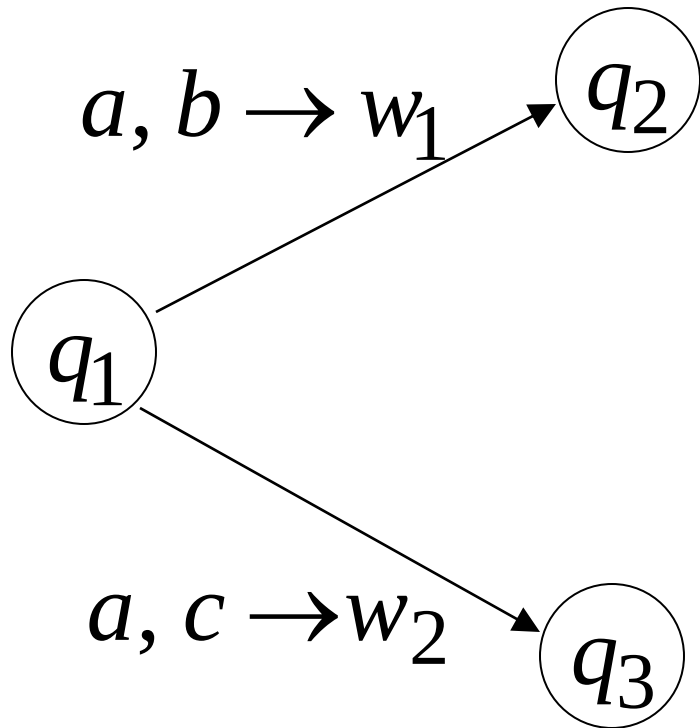
Deterministic PDA: DPDA

Allowed transitions:



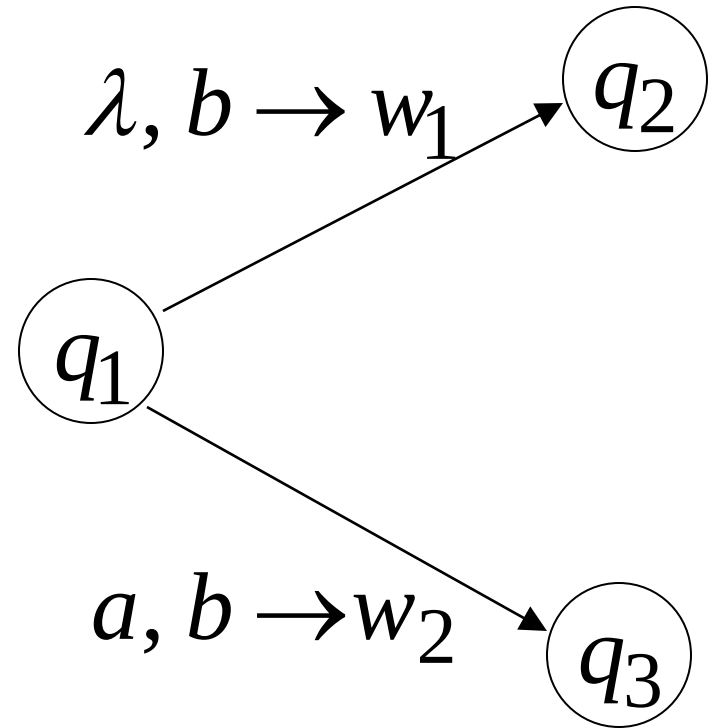
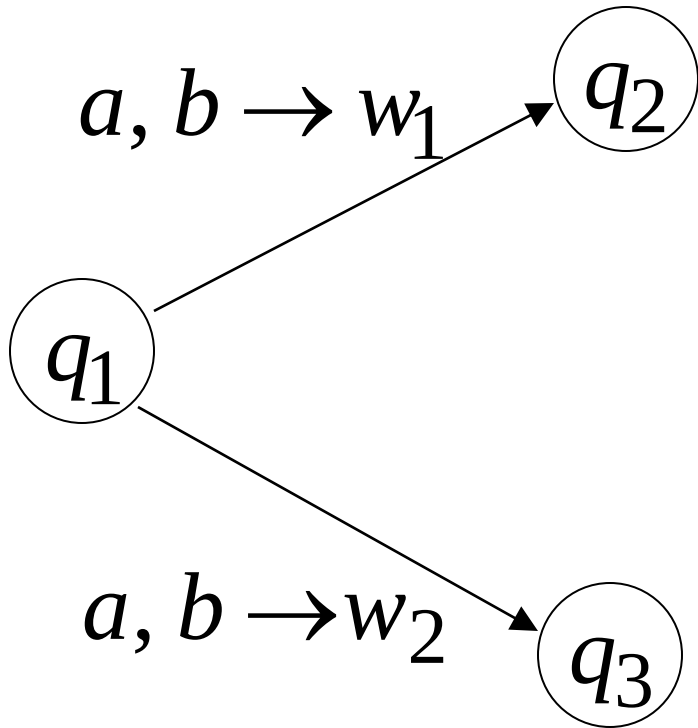
(deterministic choices)

Allowed transitions:



(deterministic choices)

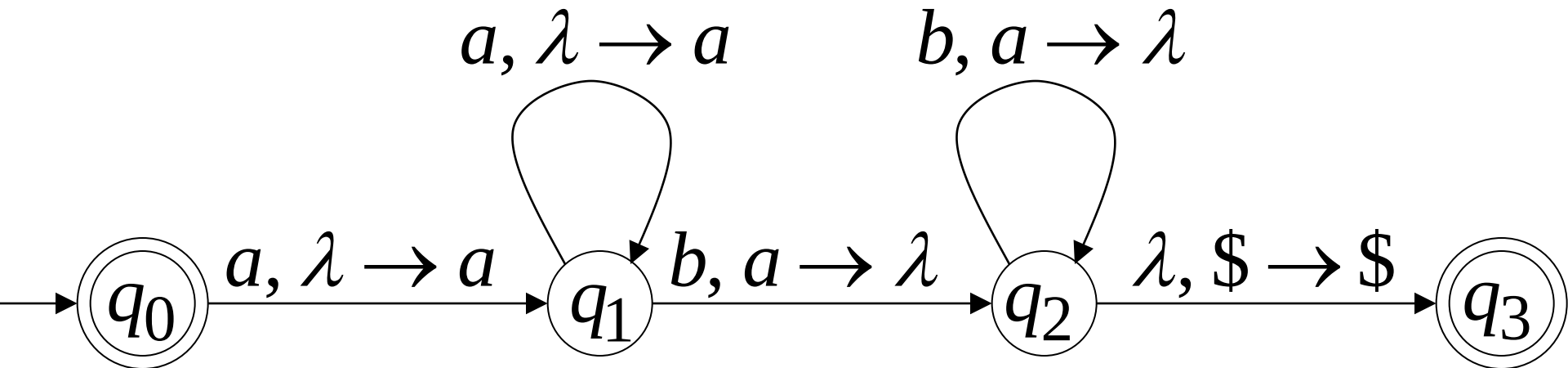
Not allowed:



(non deterministic choices)

DPDA example

$$L(M) = \{a^n b^n : n \geq 0\}$$



Definition:

A language L is **deterministic context-free** if there exists some DPDA that accepts it

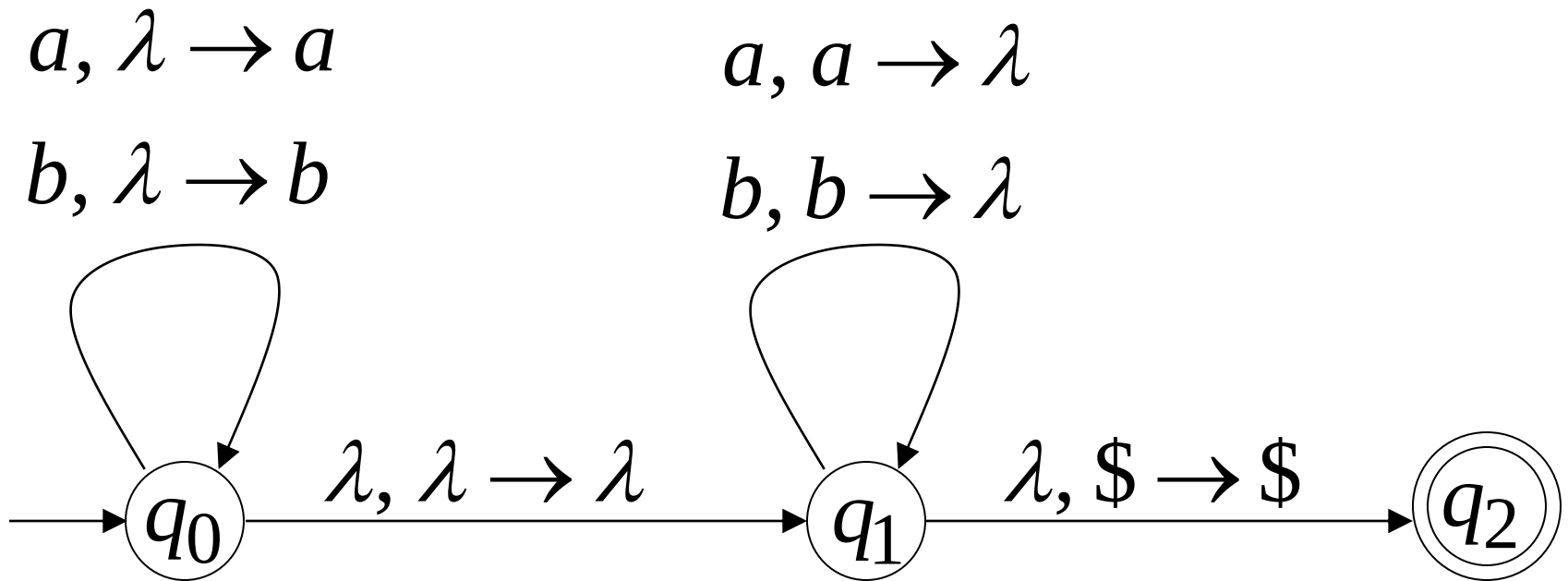
Example:

The language $L(M) = \{a^n b^n : n \geq 0\}$

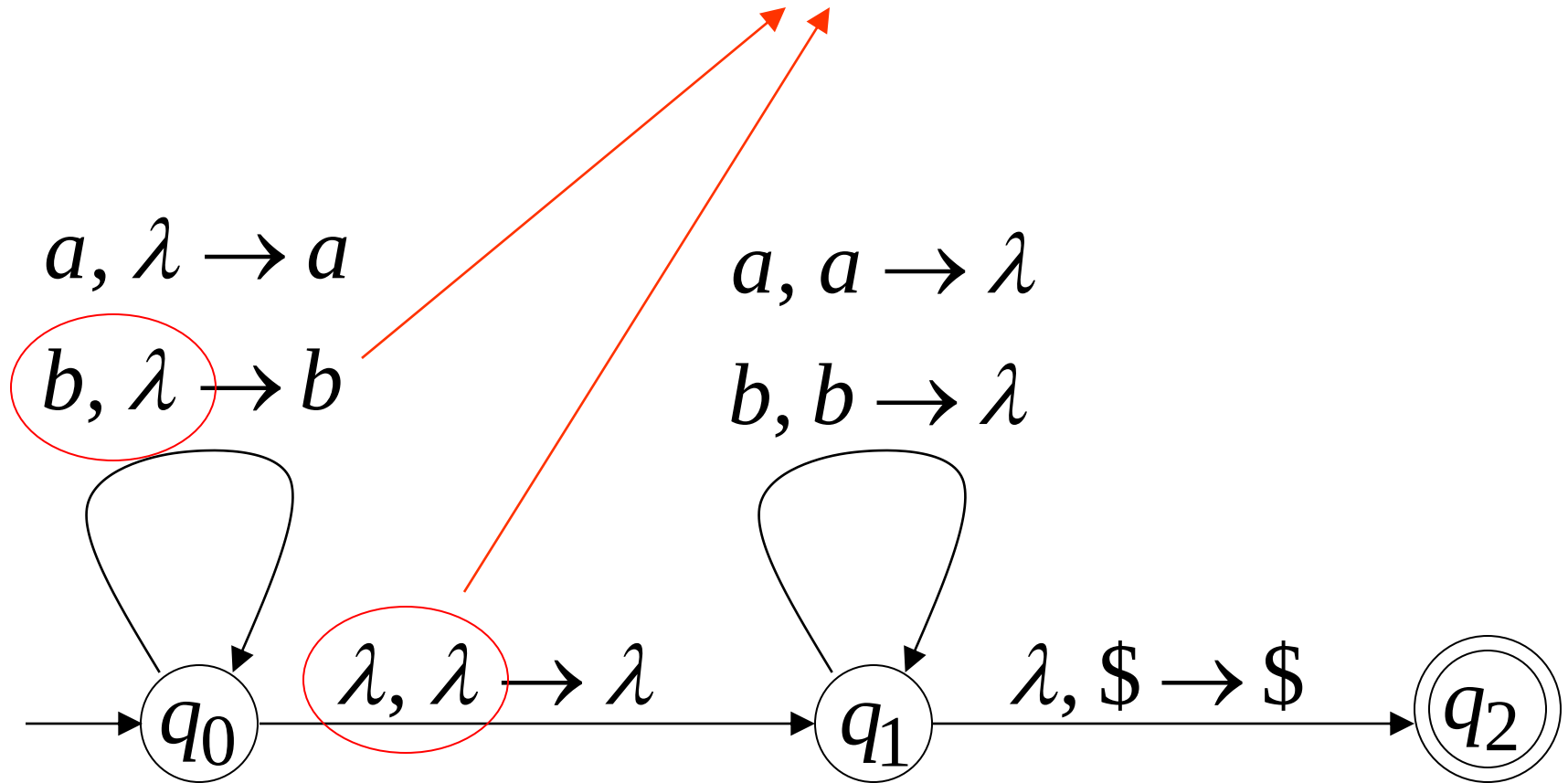
is deterministic context-free

Example of Non-DPDA (PDA)

$$L(M) = \{vv^R : v \in \{a,b\}^*\}$$



Not allowed in DPDAs



PDA_s

Have More Power than

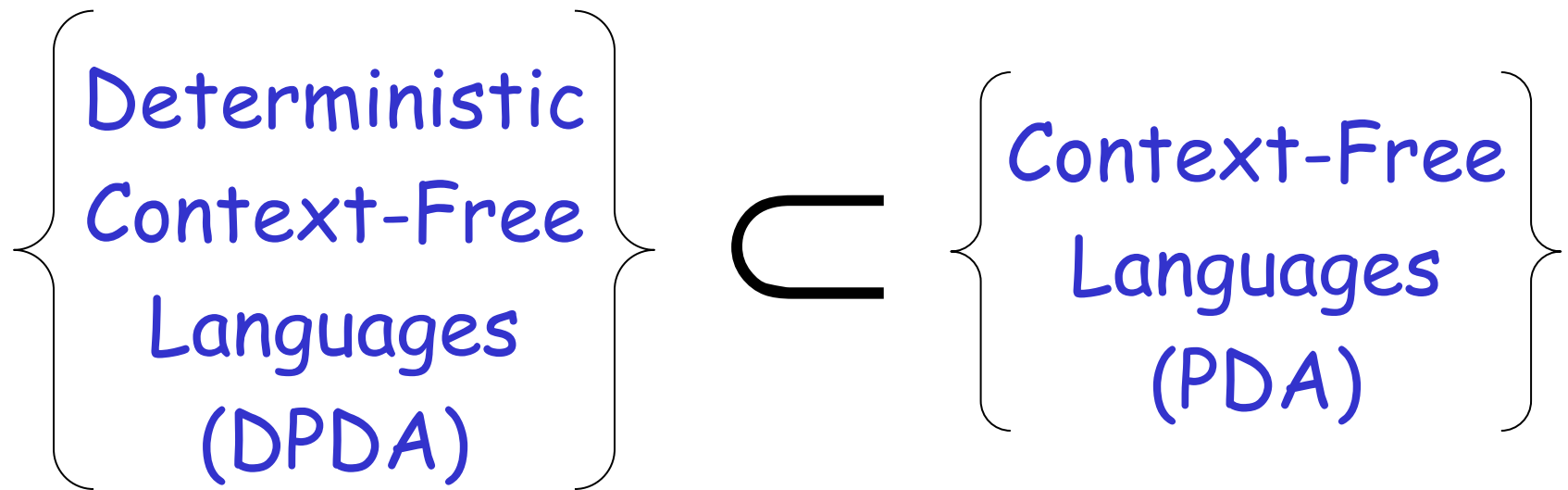
DPDA_s

It holds that:

$$\left\{ \begin{array}{l} \text{Deterministic} \\ \text{Context-Free} \\ \text{Languages} \\ \text{(DPDA)} \end{array} \right\} \subseteq \left\{ \begin{array}{l} \text{Context-Free} \\ \text{Languages} \\ \text{PDAs} \end{array} \right\}$$

Since every DPDA is also a PDA

We will actually show:



We will show that there exists
a context-free language L which is not
accepted by any DPDA

The language is:

$$L = \{a^n b^n\} \cup \{a^n b^{2n}\} \quad n \geq 0$$

We will show:

- L is context-free
- L is **not** deterministic context-free

$$L = \{a^n b^n\} \cup \{a^n b^{2n}\}$$

Language L is context-free

Context-free grammar for L :

$$S \rightarrow S_1 \mid S_2 \quad \{a^n b^n\} \cup \{a^n b^{2n}\}$$

$$S_1 \rightarrow aS_1b \mid \lambda \quad \{a^n b^n\}$$

$$S_2 \rightarrow aS_2bb \mid \lambda \quad \{a^n b^{2n}\}$$

Theorem:

The language $L = \{a^n b^n\} \cup \{a^n b^{2n}\}$

is **not** deterministic context-free

(there is **no** DPDA that accepts L)

Proof: Assume for contradiction that

$$L = \{a^n b^n\} \cup \{a^n b^{2n}\}$$

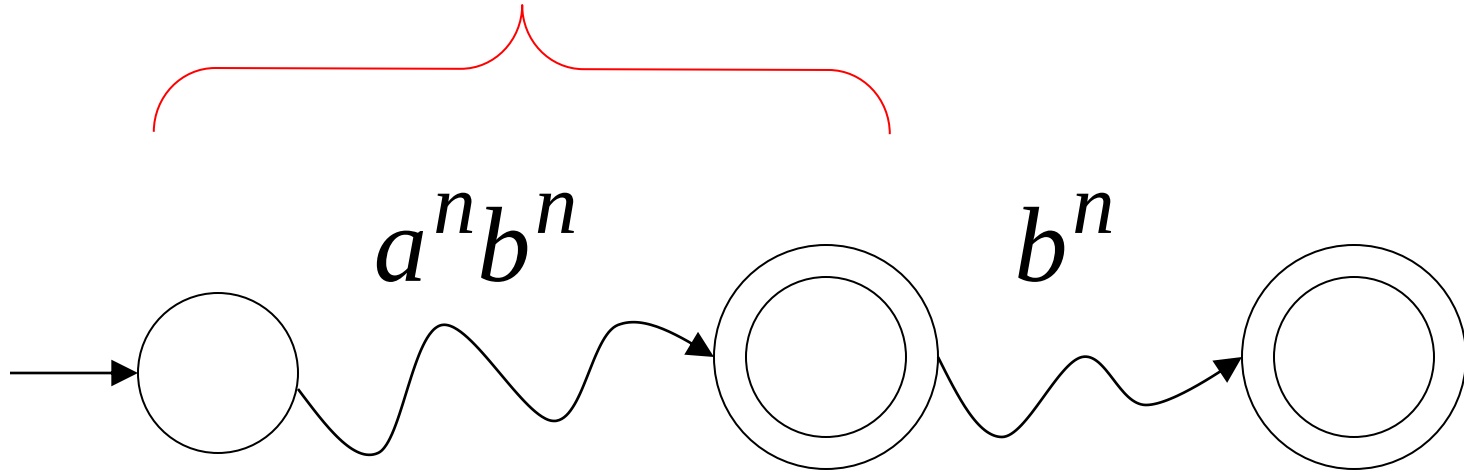
is deterministic context free

Therefore:

there is a DPDA M that accepts L

DPDA M with $L(M) = \{a^n b^n\} \cup \{a^n b^{2n}\}$

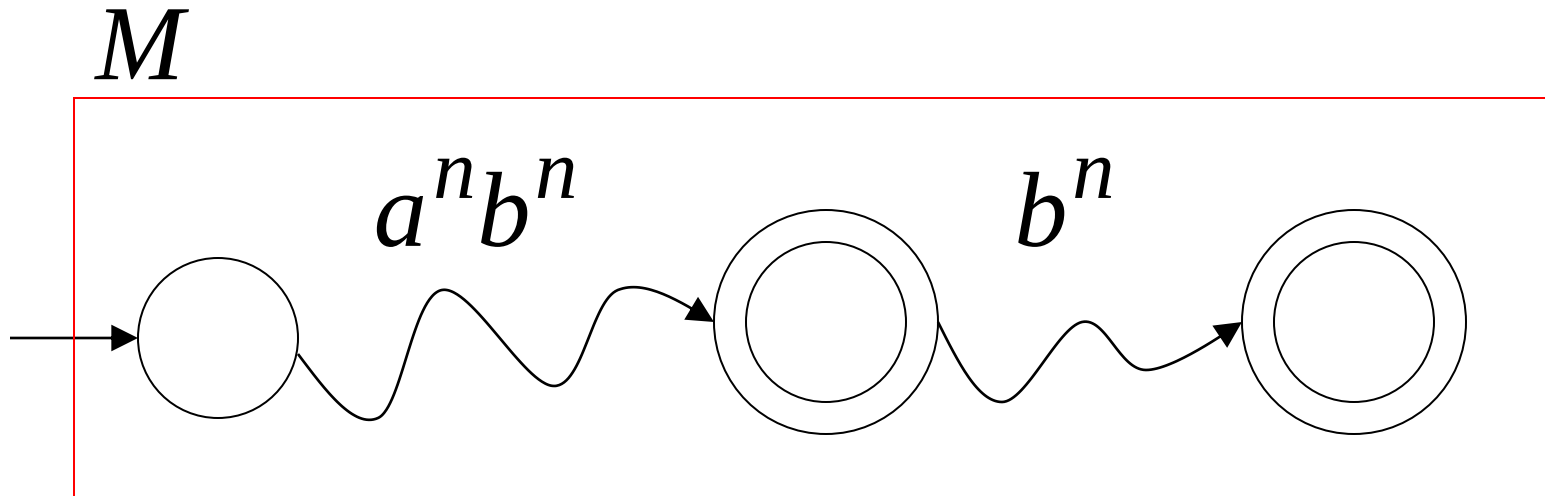
accepts $a^n b^n$



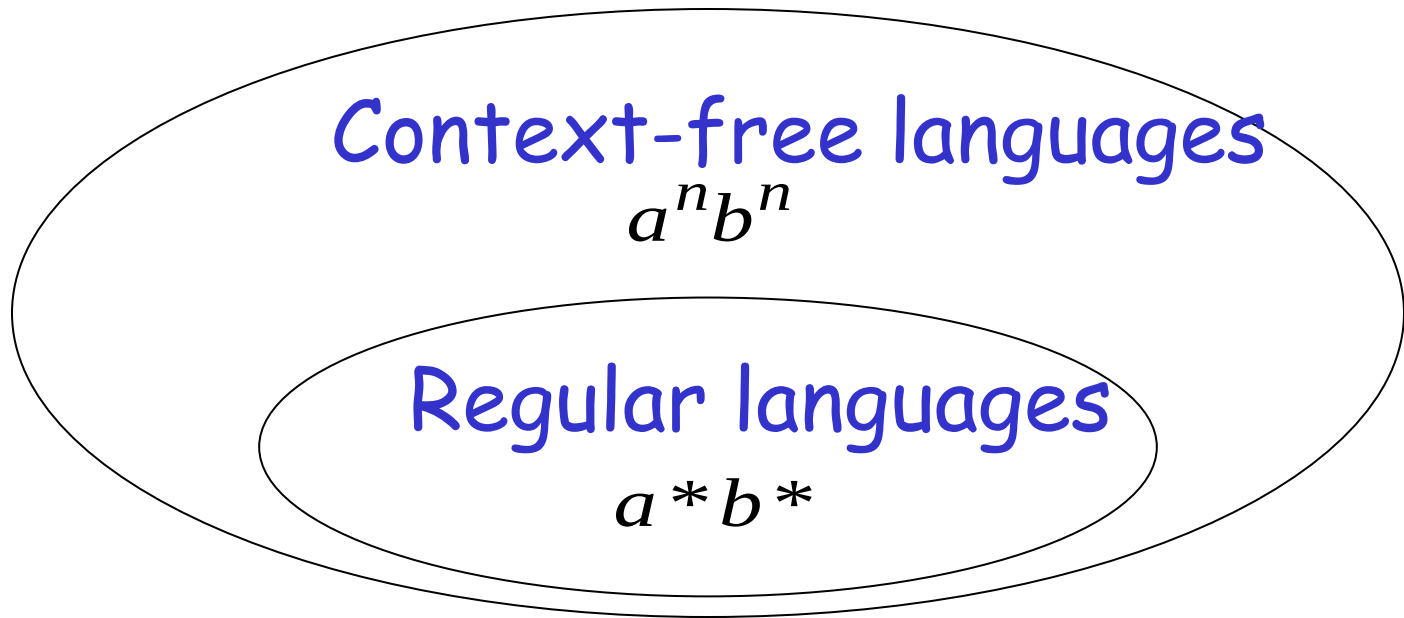
accepts $a^n b^{2n}$

DPDA M with $L(M) = \{a^n b^n\} \cup \{a^n b^{2n}\}$

Such a path exists due to determinism



Fact 1: The language $\{a^n b^n c^n\}$
is **not** context-free



(we will prove this at a later class using
pumping lemma for context-free languages)

Fact 2: The language $L \cup \{a^n b^n c^n\}$
is **not** context-free

$$(L = \{a^n b^n\} \cup \{a^n b^{2n}\})$$

(we can prove this using pumping lemma
for context-free languages)

We will construct a PDA that accepts:

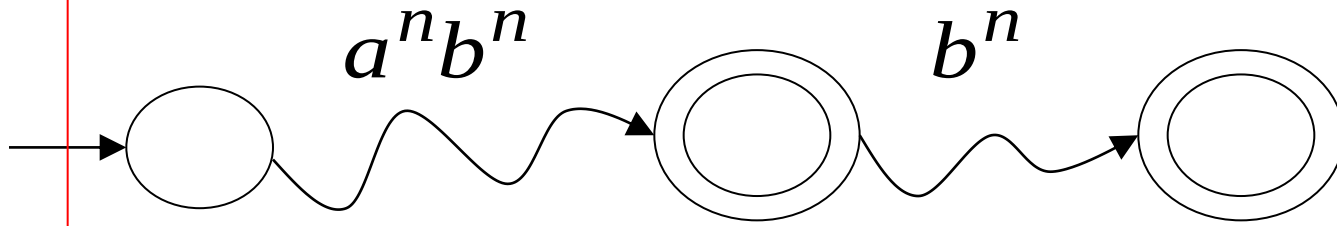
$$L \cup \{a^n b^n c^n\}$$

$$(L = \{a^n b^n\} \cup \{a^n b^{2n}\})$$

which is a contradiction!

DPDA M

$$L(M) = \{a^n b^n\} \cup \{a^n b^{2n}\}$$

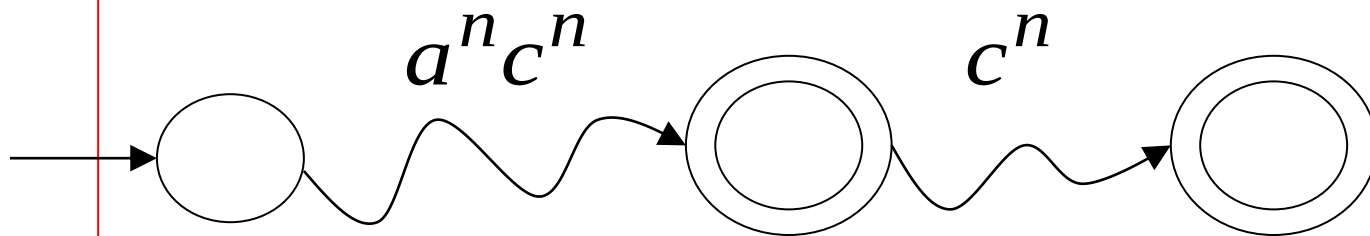


Modify M

Replace b
with c

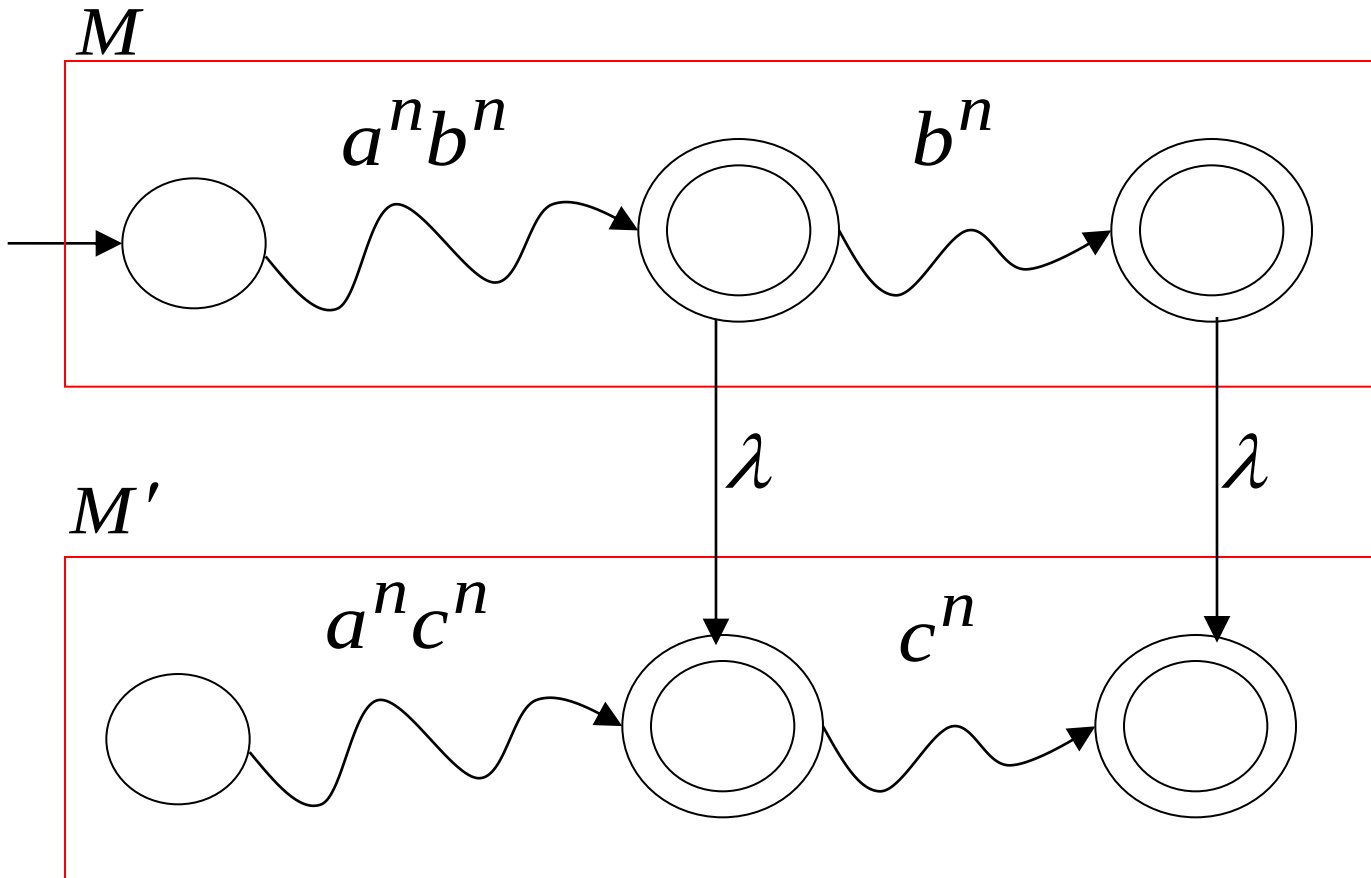
DPDA M'

$$L(M') = \{a^n c^n\} \cup \{a^n c^{2n}\}$$



A PDA that accepts $L \cup \{a^n b^n c^n\}$

Connect the final states of M
with the final states of M'



Since $L \cup \{a^n b^n c^n\}$ is accepted by a PDA
it is context-free

Contradiction!

(since $L \cup \{a^n b^n c^n\}$ is not context-free)

Therefore:

$$L = \{a^n b^n\} \cup \{a^n b^{2n}\}$$

Is not deterministic context free

There is no DPDA that accepts it

End of Proof